

■ ML & DL PERFORMANCE METRICS + OPTIMIZERS CHEATSHEET

I. PERFORMANCE METRICS — MACHINE LEARNING & DEEP LEARNING

Metrics are the compass. They tell you whether your model is learning something valuable or just burning compute cycles.

■ A. Classification Metrics

Used when the output is categorical (e.g., spam/not spam, churn/no churn).

Accuracy: For balanced datasets. Fraction of correct predictions.

Precision: Use when False Positives are costly (fraud detection, spam). Measures how many predicted positives are correct.

Recall (Sensitivity): Use when False Negatives are costly (cancer detection). Measures how many actual positives were detected.

F1-Score: Best for imbalanced data. Harmonic mean of Precision & Recall.

AUC-ROC: Binary classification, imbalanced data. Measures class separability.

Log Loss: For probabilistic classifiers. Penalizes overconfident wrong predictions.

Confusion Matrix: Visualizes TP, FP, TN, FN.

Top-K Accuracy: For multi-class problems (e.g., ImageNet).

■ B. Regression Metrics

Used for continuous outputs (e.g., price, risk score).

MAE: Simple, interpretable. Average absolute error.

MSE: Penalizes large errors. Smooth gradients.

RMSE: Same unit as target. Penalizes big errors more.

R²: Relative performance vs mean.

MAPE: Forecasting. Error in % terms.

Huber Loss: Robust to outliers.

■ C. Clustering Metrics

Used for unsupervised models.

Silhouette Score: Cluster separation (-1 to +1).

Davies–Bouldin Index: Lower = better clusters.

Calinski–Harabasz Index: Higher = better clusters.

ARI: Compares predicted vs true clusters.

■ D. NLP Metrics

Used for translation, summarization, and text models.

BLEU: n-gram overlap with reference (translation).

ROUGE: Recall-based overlap (summarization).

METEOR: Considers synonyms and stem matches.

Perplexity: Lower = better model confidence.

F1 / Accuracy: For text classification.

■ E. Computer Vision Metrics

IoU: Object detection/segmentation. Overlap of predicted vs actual area.

mAP: Detection benchmarks (COCO, Pascal).

Dice Coefficient: For medical segmentation. Measures pixel overlap.

■ F. Time Series Metrics

MAE / RMSE: Standard forecasting metrics.

MAPE / SMAPE: Business metrics in % terms.

MASE: Scalable across datasets.

DTW: Measures temporal sequence similarity.

■■ II. OPTIMIZERS IN DEEP LEARNING

Optimizers decide how weights get updated to minimize loss functions.

SGD: Simple, reliable. Small models. Slow convergence.

Momentum: Adds velocity. Faster learning.

NAG: 'Look ahead' momentum. Accurate but complex.

Adagrad: Adaptive lr per parameter. Sparse data.

RMSProp: Great for RNNs and sequential data.

Adam: Universal optimizer. Combines momentum & RMSProp.

AdamW: Adam + weight decay (transformers).

Nadam: Adam + Nesterov. Fast but heavy.

Adadelta: Online/streaming data. No manual lr.

LAMB / Lion / RAdam / AdaBelief: For large-scale models (BERT, GPT).

■ TL;DR — QUICK CHEAT SHEET

Binary Classification: Accuracy, F1, ROC-AUC → Adam, SGD+Momentum

Multi-class: Accuracy, F1-macro → Adam, NAG

Regression: RMSE, R^2 → Adam, RMSProp

Clustering: Silhouette, ARI

NLP: BLEU, ROUGE, Perplexity → AdamW, Nadam

CV: IoU, mAP → Adam, SGD+Momentum

Time Series: MAPE, RMSE → Adam, RMSProp